



Department of Computer Science and Business Systems

Academic Year 2022– 2023(Odd Semester)

Degree, Semester & Branch: B.Tech – I Semester CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Dr.S.Erana Veerappa Dinesh, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Guess an Integer number in a range
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Role play
- **Justification:**

A number guessing game is a simple guessing game where a user is supposed to guess a number between 0 and N in a maximum of 3 attempts. The game will end after 3 attempts and if the player failed to guess the number, and then he loses the game. This active learning strategy is chosen to help students to understand the concept of implementing an algorithm for guessing an integer problem.

- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - Faculty explained the concept of the problem. Two students were asked to come in front of the classroom.
 - Student 1 was asked to fix a random number 'x' in the range of 1 to 50
 - Student 2 was asked to guess the number 'y'. If the guessed number 'y' is greater than the random number 'x' then student 1 says "Too high".
 - If the guessed number 'y' is less than the random number 'x', then student 1 says "Too low"
 - If the guessed number 'y' is equal to the random number 'x', then the student 1 says "You win"
 - To make the activity more interesting, number of guesses is fixed. If the student 2 fails to guess a number within the fixed number of guesses, the student 1 says "You lose"
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)



- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	It involves logical thinking and make the students able to think about problem solving approach	Students will be able to understand the problem in real manner than theoretical explanation	Students will be able to provide solution for the problem by writing python code through this activity	As the students are involved in the activity as team of two, the team work skill improved	Students communication skill will be improved as they are facing audience and communicating the activity	The problem solving skill earned through this activity helps the students in solving various problems

- Images / Screenshot of the practice:



Fig 1: Students involvement in Number Guessing Game

- Reflective Critique:

- ❖ *Feedback of practice from students and other stakeholders:*

- Students said it is quiet easy to understand when compare with other illustrative programs
- Through this activity the students felt easy to write the program

- ❖ *Benefit of the practice:*



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- Students understanding of the problem is improved.
- Students coding skill through this activity is better.

❖ **Challenges faced in implementation:**

- This activity involves only 2 members.
- Unable to make all the students to involve in this activity.

References:

- ❖ <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS8493-OS-Roleplay.pdf>
- ❖ https://aminoapps.com/c/attack-on-titan/page/blog/different-types-of-roleplay-lit-team/YoYg_nMTbux63E26M7z6xpRxkovvoBXPj

Signature of Faculty Member

HOD